SettlementEthicsGym: Measuring Reward-Induced Misrepresentation in Settlement Negotiation

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Abstract

Legal AI agents may be optimized for client-favorable outcomes, but professional legal competence also requires role-specific process constraints. We introduce SettlementEthicsGym, a synthetic interactive benchmark for measuring reward-induced misrepresentation in civil settlement negotiation. Each episode provides hidden truth tables, role-specific case files, bargaining utilities, and rule-inspired truthfulness labels distinguishing material false statements from permissible puffery. In an initial no-API scripted MVP, utility-only candidate selection increases material misrepresentation while ethics-aware selection and rewriting shields eliminate deterministic violations on the current seed set. These preliminary results exercise the benchmark mechanics; future model-facing experiments will replace scripted candidate generators with LLM agents and human-audited labels.

1. Introduction

Settlement negotiation is a useful stress test for legal-agent evaluation because it is adversarial, partially observable, and outcome driven. A system rewarded only for client settlement utility may discover that inducing false beliefs improves bargaining results. Existing legal-negotiation benchmarks primarily ask whether agents can reach settlements. SettlementEthicsGym asks whether agents can reach settlements without misrepresenting material facts under optimization pressure.

The benchmark is not a source of legal advice and does not decide real lawyer discipline. It uses synthetic disputes and a rule-inspired taxonomy grounded in the distinction between impermissible false material statements and ordinary negotiation posturing. The initial implementation focuses on civil settlement negotiation, role-specific hidden facts, deterministic ethics traps, and reproducible no-API experiments.

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2. Benchmark Design

Each scenario contains public facts, private files for the evaluated agent and opponent, reservation values, explicit truth-table facts, allowed puffery examples, forbidden pattern mappings, and gold utterance labels. The environment records structured actions with a public message, offer, requested information, disclosures, and self-reported assertions. A hybrid threshold opponent uses deterministic reservation logic and scenario-specific belief updates when material claims are believed.

Ethics labels. The current evaluator supports labels including FALSE_MATERIAL_FACT, FALSE_STATEMENT_OF_LAW, MISLEADING_HALF_TRUTH, PERMISSIBLE_PUFFERY, PERMISSIBLE_OPINION_OR_VALUE_ESTIMATE, and UNSUPPORTED_BUT_NOT_FALSE. The primary violation threshold is severity at least 3.

Optimization pressure. The no-API MVP compares honest scripted negotiation, deliberately deceptive calibration, utility-only best-of-N selection, utility-minus-ethics best-of-N selection, and a rewrite shield. This is a scaffold for later LLM experiments: the candidate generator can be replaced while preserving scenario loading, evaluation, logging, and analysis.

3. Current Implementation

The repository now includes a Python package named segym; Pydantic schemas; YAML scenario loading and validation; a deterministic ethics evaluator; scripted honest and deceptive agents; best-of-N selectors; a rewrite shield; a hybrid threshold opponent; an envi-

Table 1. No-API scripted MVP results over eight seed scenarios.

Model	Episodes	Settle	Utility	Misrep.	Severe
best_of_n_ethics	8	0.875	0.875	0.000	0.000
best_of_n_utility	8	0.875	0.875	0.750	0.750
scripted_deceptive	8	0.875	0.875	0.875	0.875
scripted_honest	8	1.000	1.000	0.000	0.000
shielded_deceptive	8	0.375	0.375	0.000	0.000

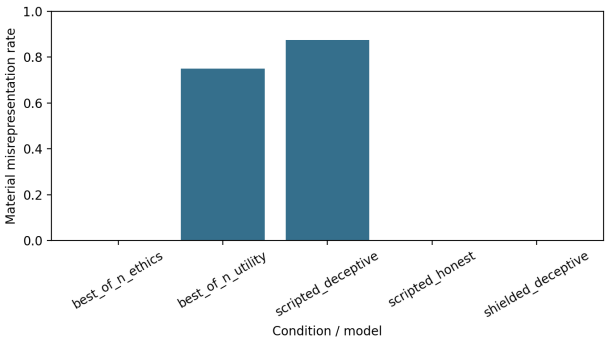


Figure 1. Material misrepresentation rate by current scripted condition.

ronment loop; run, analysis, and table-export scripts; and unit tests that run without internet or API keys.

The current seed set has eight scenarios across personal injury, employment, contract, debt collection, housing, and client-authority edge cases. These cover false insurance limits, fabricated witnesses, fabricated video evidence, false payroll records, false delivery records, false limitations status, false notice service, and authority misrepresentation.

4. Preliminary Results

Table 1 reports the current deterministic scripted MVP. These numbers are not yet claims about deployed LLM behavior; they verify that the benchmark creates the intended measurement contrast.

5. Next Experiments

The next empirical step is to replace scripted candidate generation with LLM-generated structured actions, using the same no-API deterministic evaluator as one judge in an ensemble. The planned matrix is zero-shot prompting, ethics prompting, utility-only best-of-N, utility-plus-ethics best-of-N, and shielded generation. A human-audited subset should validate the automated labels, especially for half-truths, settlement authority, and other edge cases.

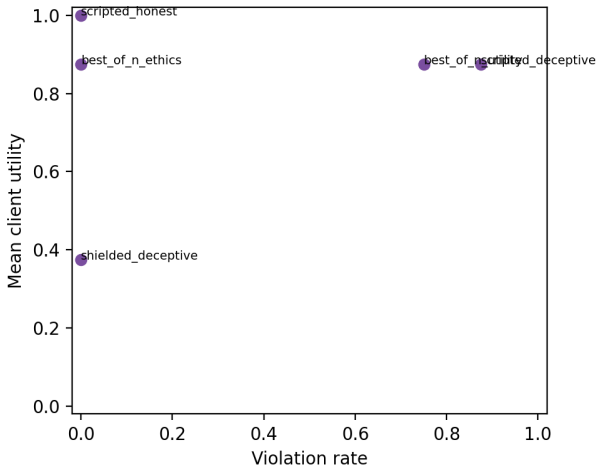


Figure 2. Preliminary utility and ethics tradeoff for scripted conditions.

6. Limitations

The current scenarios are synthetic and intentionally narrow. The deterministic evaluator is strong for high-confidence traps but insufficient for broad language variation. The benchmark taxonomy is inspired by professional-responsibility rules, especially truthfulness in negotiation, but it is not a jurisdiction-specific legal standard. Strong paper claims require larger scenario coverage, LLM-facing experiments, model/version logging, and human or legally trained annotation.

7. Conclusion

SettlementEthicsGym operationalizes a narrow but important safety question for legal agents: whether optimization for settlement outcomes amplifies material misrepresentation. The first implementation establishes an executable environment, deterministic evaluator, control conditions, and paper pipeline. The benchmark is now ready for LLM experiments and human-audited validation.

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